

Decisions, Debugging & Loops

Post-Assessment 2 · 15 questions · 49 points

Name: _____

Date: _____

Score: / 49

This is a **post-assessment**. It shows what you learned in this unit. Try every question, and check your answers if you have time.

Work on your own, without a computer. You have about **45 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

1. True or False?

4 points

Write **True** or **False** for each comparison.

Comparison	Result	Comparison	Result
<code>2 >= 9</code>		<code>3 != 8</code>	
<code>"dog" == "dog"</code>		<code>6 < 6</code>	

2. One equals or two?

2 points

(a) Which line has a bug? Circle one. (1 pt)

- A

`lives = 3`

B

`if lives = 0:`
- C

`if lives == 0:`

D

`lives = lives - 1`

(b) In your own words, what does `==` do? (1 pt)

3. Can You Ride?

4 points

```
if height >= 140:
    print("Big coaster")
elif height >= 120:
    print("Kiddie coaster")
elif height >= 100:
    print("Carousel")
else:
    print("Maybe next year")
```

What does the code print when height has each value below?

height	Prints
125	
140	
99	
100	

4. Beach Day

4 points

```
if temp > 25 and not raining:
    print("Go swimming")
if temp < 10 or raining:
    print("Stay inside")
if temp > 20:
    if raining:
        print("Bring an umbrella")
    else:
        print("Wear sunscreen")
```

(a) What prints when temp = 30 and raining = False?

(2 pts)

(b) What prints when temp = 22 and raining = True?

(2 pts)

5. Name that error

4 points

Each program crashes. Write the error name: **SyntaxError**, **NameError**, **TypeError**, or **IndentationError**. Each one is used once.

Code	Error
<code>total = 5 + "5"</code>	
<code>lives = 3 while lives > 0: print("alive")</code>	
<code>lives = 3 if lives > 0 print("alive")</code>	
<code>Print("hi")</code>	

6. Bug Zoo: Movie Vote

3 points

This program has **three** bugs. Circle each bug and write the fixed program in the box.

```
votes = int(input("Votes for comedy: "))
if votes > 5:
    print("Comedy night!")
elif votes == 5
    print("It's a tie!")
else:
    print("Pick another movie")
```


7. Flipping coins

2 points

```
import random
number = random.randint(1, 10)
side = random.choice(["heads", "tails"])
```

(a) Which numbers can `number` be? Circle one.

(1 pt)

☐ (A) 0 through 9

☐ (B) 1 through 10

☐ (C) 1 through 9

☐ (D) 0 through 10

(b) Name every value `side` could be.

(1 pt)

8. Triple Jump

3 points

Write exactly what this program prints.

```
tries = 1
while tries < 50:
    print(tries)
    tries = tries * 3
print("Done")
```

9. Stop that loop!

2 points

```
lives = 3
while lives > 0:
    print("Keep playing")
    lives = lives + 1
```

(a) What goes wrong when this runs?

(1 pt)

(b) Rewrite the one line that needs to change so it prints Keep playing 3 times.

(1 pt)

10. Which loop?

2 points

For each job, write **for** or **while**, and give a short reason.

(a) Keep rolling a die until you roll a 6.

(1 pt)

(b) Print each letter of the word "python".

(1 pt)

11. What does range() give?

4 points

List every number each range produces, in order.

Code	Numbers
range(6)	
range(3, 8)	
range(0, 20, 5)	
range(8, 0, -2)	

12. Counting and adding

4 points

(a) What does this print?

(2 pts)

```
word = "mississippi"
count = 0
for letter in word:
    if letter == "s":
        count = count + 1
print(count)
```

(b) What does this print?

(2 pts)

```
total = 0
for i in range(3, 7):
    total = total + i
print(total)
```

13. Skip and stop

3 points

Write exactly what this program prints.

```
for n in range(2, 12):  
    if n == 4:  
        continue  
    if n == 7:  
        break  
    print(n)
```

14. Upside-down Triangle

3 points

Write exactly what this program prints.

```
for row in range(3, 0, -1):  
    line = ""  
    for col in range(row):  
        line = line + "#"  
    print(line)
```

15. Guess the Number

5 points

Write a program where the computer picks a secret number from 1 to 20 with `random.randint`, and the user keeps guessing until they get it.

- After a wrong guess, print `Too high` or `Too low`.
- When they get it, print `You got it in 4 guesses!` (with the real number of guesses).

END OF ASSESSMENT